

Pasquinelli Lab's Own Narrow AI LLM:

An AI Research Agent for miRNA Biology,
Aging & RNA Processing

Built on Karpathy's autoresearch framework • Amy_Autoresearch branch
github.com/jlpfly/Karpathy_autoresearch • Dell XPS 15 (RTX 3050 Ti)
Training corpus: 25 years of Pasquinelli lab publications + the full miRNA field



Autoresearch • RAG • Fine-Tuning • Autonomous Agent Loop • miRNA Biology • C. elegans Aging

Imagine an AI that has read every paper your lab has ever published specifically trained on your area and corresponding field

What this LLM knows

- Every Pasquinelli lab paper from let-7 (2000) through the 2024 aging RNA-seq study
- Full miRNA field: TargetScan, miRDB, AGO biology, CLIP/CLASH datasets
- C. elegans aging literature, heat-shock recovery, poly(A) tail biology
- Unpublished datasets, protocols, and experimental notes (optional)

What this means for the lab

"What are the strongest open questions in let-7 biology right now?"

→ Amy, strategic planning

"Design a follow-up experiment for our miR-238 longevity finding."

→ Postdoc, experimental design

"Explain the ALG-1 vs ALG-2 aging phenotype to me."

→ Rotation student, onboarding

"What gaps in the aging-miRNA field should our R01 address?"

→ Grant writing, Significance section

This is not a generic chatbot. It is trained specifically on the body of work most relevant to your research program.

Three layers make this agent work: Corpus • Model • Agent Loop

LAYER 1



The Corpus

What it reads

- All Pasquinelli lab papers (2000–2025): ~55 papers
- Field-defining papers: Bartel, Ambros, Tuschl labs
- PubMed abstracts, PMC full text, GEO metadata
- Lab protocols, grant aims, meeting notes (optional)

LAYER 2



The Model

How it learns

- Base: Llama 3, Mistral, or GPT-4 via API
- Method: RAG or supervised fine-tuning on corpus
- Hardware: RTX 3050 Ti (already in the lab)
- Time to first version: 1–2 days setup + overnight

LAYER 3



The Agent Loop

How it acts

- Reads program.md (your research strategy)
- Iterates: reads → extracts → hypothesizes → ranks
- Outputs: tables, hypothesis boards, reading queues
- Human stays in control: review, redirect, refine

Building the Pasquinelli Lab LLM takes one weekend, not one year

01

Set up the environment

Day 1 · ~2 hours

```
git clone
https://github.com/jlpfly/
Karpathy_autoresearch
git checkout Amy_Autoresearch
uv python install 3.11
uv sync
uv run prepare.py
```

Hardware: Dell XPS 15 with RTX
3050 Ti
— already validated

02

Build the corpus

Day 1 · ~3 hours

Export PDFs of all Pasquinelli
lab
papers from PubMed/PMC

Add key field papers (Bartel,
Ambros,
let-7 foundational work)

pdftotext paper.pdf paper.txt

Place in corpus/ directory

03

Configure the agent

Day 1 · ~1 hour

Edit program.md to define the
agent's role

Set focus: "You are a research
assistant for the Pasquinelli
Lab..."

Define output format: tables,
ranked lists, hypothesis boards

04

Run overnight

Day 2

```
uv run train.py
# 5-minute baseline validation
```

Launch Claude/Codex with
program.md — let it iterate

Wake up to a log of
experiments
and structured outputs

05

Query and iterate

Ongoing

Start asking questions
(see example prompt slides)

Refine program.md based on
output quality

Add new papers as they
are published

The quality of the corpus determines the quality of the intelligence

Layer	Content	Papers / Items	Priority
Core Lab	All Pasquinelli lab publications 2000–2025	~55 papers	Essential
Field Foundation	Bartel, Ambros, Tuschl landmark miRNA papers	~30 papers	Essential
Aging & C. elegans	Kaeberlein, Murphy, Curran aging papers	~20 papers	High
Poly(A) & Translation	Richter, Gilbert, Wickens PABP/translation papers	~15 papers	High
Databases	TargetScan, miRDB, CLIP-seq datasets (metadata)	~5 sources	Medium
Lab Materials	Protocols, grant aims, meeting notes	Variable	Optional

Key principle: Start with the lab's own papers. The agent will already be more useful than a generic LLM after just this first layer.

Sources: PubMed Central ([pmc.ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov)) • Semantic Scholar API • bioRxiv preprints

Strategic questions that used to take days to answer

Grant strategy

*“Summarize the strongest open questions in miRNA-mediated aging regulation in *C. elegans* after 2020. Rank them by how well-supported they are and how testable they are with our current tools.”*

→ Output: Ranked hypothesis table with citations, evidence strength scores, and suggested experimental approaches.



Mechanism mapping

“Map the evidence for how ALG-1 and ALG-2 produce opposing aging phenotypes. What are the proposed mechanisms? What is still unresolved?”

→ Output: Mechanistic diagram in text form, evidence table, list of unresolved questions with citations.



Cross-paper synthesis

“Compare our heat-shock recovery paper (2021) with our aging RNA-seq paper (2024). Which miRNAs appear in both contexts? What does that suggest about shared regulatory programs?”

→ Output: Overlap analysis, shared gene/miRNA lists, synthesis paragraph with citations.



Field positioning

“How does our lab’s expression-not-sequence interpretation of miRNA target specificity compare to the current consensus? Who agrees, who disagrees, and what would resolve it?”

→ Output: Field map with lab positions, supporting/contradicting papers, resolution experiments.



Daily research acceleration for every lab member

Experimental design

"I'm planning a follow-up to the miR-238/239ab longevity paper. What are the three most testable hypotheses that would extend this work? What controls would be critical for each?"

Output:
Three hypothesis cards with experimental design, controls, and relevant citations.

Literature catch-up

"I just joined the lab. Give me a structured reading path from the foundational let-7 papers through the most recent aging work. Start with what I need to understand the 2024 NAR paper."

Output:
Tiered reading list with annotations, estimated reading time, key concepts per paper.

Data interpretation

"We found that transcript X has a short poly(A) tail but high translation efficiency. Is this an outlier? What mechanisms could explain it based on the PABP literature?"

Output:
Literature-grounded interpretation, candidate mechanisms, papers to read.

Manuscript prep

"Check the consistency between our methods section and figure legends for the heat-shock recovery paper. Flag any discrepancies in sample sizes, strain names, or statistical tests."

Output:
Discrepancy table with line numbers and suggested corrections.

Journal club prep

"Summarize this new paper on miRNA-mediated translational repression in 5 points. How does it relate to our lab's work? What questions should we ask the authors?"

Output:
Summary, relevance assessment, 3–5 discussion questions.

Let the agent work while you sleep — wake up to structured results

Task 1: Literature sweep

“Search for all papers published in the last 6 months on miRNA regulation in aging organisms. Summarize each, score relevance to our lab’s work (1–10), and flag any that contradict our published findings.”

→ By morning: Ranked paper list • Relevance scores • Contradiction flags

Task 2: Cross-dataset replication

*“Download all public *C. elegans* aging RNA-seq datasets from GEO. Score each using the gene signatures from our 2024 NAR paper. Report which signatures replicate and which do not.”*

→ By morning: Replication score table • Forest plots • Core aging marker gene set

Task 3: Target concordance

*“For *let-7* and *miR-238*, pull predictions from TargetScan and miRDB. Cross-reference against our published functional data. Build a concordance score for each predicted target.”*

→ By morning: Concordance-scored target table • Venn diagrams • High-confidence target shortlist

Task 4: Hypothesis generation

“Starting from our 2021 heat-shock recovery paper, find all related papers on miRNA-mediated stress recovery. Generate 10 testable hypotheses ranked by evidence strength and experimental feasibility.”

→ By morning: Ranked hypothesis board • Citations • Experimental sketches

Five experiments the agent can run on real datasets — not just literature

01

Age-Dependent Isoform Shift Quantification

Reanalyzes 2024 direct RNA-seq aging data to build age-stratified isoform usage matrix.

Anchor:

2024 NAR aging direct RNA-seq (PMC11662692)

Outputs:

Ranked switching gene list
Age-group heatmaps
GO enrichment

02

Cross-Dataset miRNA Target Concordance

Integrates TargetScan, miRDB, and CLIP/CLASH; cross-references against lab functional data.

Anchor:

2023 miR-238/239ab paper (PLoS Genetics)

Outputs:

Concordance-scored target table
Venn diagrams
Top 20 targets per miRNA

03

Poly(A)-Tail & Translation Efficiency Integration

Correlates poly(A)-tail length with translational output; identifies outlier transcripts.

Anchor:

2022 NAR PABP paper (PMC9071453)

Outputs:

Tail length vs. TE scatter plots
Outlier transcript lists
GO enrichment

04

Stress-Recovery Time-Series Gene Module Detection

Clusters genes into co-regulated modules from heat-shock recovery dynamics; maps miRNA targets.

Anchor:

2021 PLoS Genetics heat-shock (PMC8370650)

Outputs:

Temporal profile plots
miRNA-module enrichment scores
Network visualizations

05

Automated Cross-Dataset Replication

Downloads public C. elegans aging RNA-seq datasets from GEO; scores each with lab-derived signatures.

Anchor:

2024 NAR aging direct RNA-seq (PMC11662692)

Outputs:

Replication score table
Forest plots
Core aging marker gene set

Each experiment feeds the next — a self-reinforcing evidence network

#1 Isoform Shifts

feeds isoform-switching gene lists into →

2024 NAR aging RNA-seq



#2 Target Concordance

provides curated miRNA target lists for →

2023 miR-238/239ab paper



#3 Poly(A) / Translation

identifies tail-length outliers, checked by →

2022 NAR PABP paper



#4 Recovery Modules

maps miRNAs to recovery programs, validated by →

2021 PLoS Genetics heat-shock



#5 Cross-Dataset Replication

validates everything across independent data

2024 NAR aging RNA-seq



Running all five creates a compounding evidence network. The agent can run all five overnight and synthesize the results by morning.

Autonomous AI research is no longer speculative — it is happening now

System	Organization	Milestone	Relevance to Lab
autoresearch	Karpathy (2026)	Agent-driven code iteration on consumer GPU; adopted worldwide in weeks	<i>Foundation of this repo</i>
PaperQA2	FutureHouse	Beats PhD researchers at biology literature search (LitQA2 benchmark)	<i>Direct analog for miRNA literature work</i>
ContraCrow	FutureHouse	Finds ~2.34 contradicted statements per biology paper	<i>Contradiction finder for miRNA field</i>
WikiCrow	FutureHouse	Gene-level summaries from 1 million papers, more accurate than Wikipedia	<i>Gene-level knowledge base for C. elegans</i>
AI Scientist v2	Sakana AI	First AI-generated paper to pass peer review at ICLR workshop	<i>Full autonomous research loop</i>
Coscientist	CMU (Gomes lab)	Autonomous chemistry experiments via robotic API (Nature, 2023)	<i>Closes hypothesis-to-wet-lab loop</i>

The question is not whether autonomous research tools will be useful in biology labs — it is how quickly labs like the Pasquinelli lab adopt them.

Two paths to a lab-specific LLM — choose based on your goals

Dimension	RAG (Retrieval-Augmented Generation)	Fine-Tuning
Setup time	Hours to days	Days to weeks
Hardware needed	CPU or small GPU	GPU (RTX 3050 Ti sufficient)
Corpus update	Add files, re-index (minutes)	Retrain (hours)
Answers cite sources	Yes, automatically	Requires prompting
Best for	Literature Q&A, hypothesis generation	Style adaptation, specialized reasoning
Cost	Near-zero (local) or low API cost	Near-zero (local)
Recommended for lab	START HERE →	Add later for specialized tasks

RAG in plain English: The model looks up relevant passages from your corpus before answering — like giving the LLM a searchable library of your papers.

Fine-tuning in plain English: You train the model's weights on your corpus so the knowledge is baked in. More powerful but more effort to update.

program.md is what makes this a Pasquinelli Lab agent, not a generic chatbot

You are a research assistant for the Pasquinelli Lab at UC San Diego.

Your expertise covers:

- microRNA biology, biogenesis, and target regulation
- C. elegans genetics, aging, and stress response
- Poly(A) tail biology and translational regulation
- let-7 and small RNA regulatory mechanisms
- RNA processing and isoform regulation

Your corpus includes all Pasquinelli lab publications from 2000 to 2025, plus key papers from the Bartel, Ambros, Richter, and Kaeberlein labs.

For each task:

1. Search the corpus for relevant evidence
2. Extract key claims with citations
3. Grade evidence: strong / moderate / weak / contested
4. Identify gaps and open questions
5. Output as structured tables and ranked lists

Current focus: [INSERT CURRENT RESEARCH QUESTION]

Why this file matters



The program.md file is your research strategy. It tells the agent:

- What domain it is working in
- What corpus to draw from
- How to format outputs
- What the current research focus is

Invest time in refining it. Each iteration makes the agent more useful.

Template available in:

[amy_autoresearch_howto/](#)
[Amy_Autoresearch branch](#)

You can have a working lab LLM agent in 48 hours

This week (hours to days)

1. Literature gap-finder for next grant proposal

1–2 days

2. New-paper alert and summary service

Half a day

3. Manuscript consistency checker

Hours per manuscript

4. Hypothesis ranking board from a seed paper

1–2 days

5. Lab knowledge base for onboarding

Ongoing, seed in hours

This month (days to weeks)

1. Full corpus build: all lab papers + key field papers

2. RAG-based query system: ask questions, get cited answers

3. Isoform shift analysis on 2024 aging data

4. Cross-dataset replication of aging signatures

5. Living hypothesis board that updates as new papers appear

All how-to guides, starter code, and program.md templates: [amy_autoresearch_howto/](#) • [github.com/jlpfly/Karpathy_autoresearch](#)

The Pasquinelli Lab is positioned to lead in AI-augmented RNA biology

Why this lab, why now

- 25 years of publications = a rich, coherent training corpus
- Deep expertise in a focused domain = high-quality agent outputs
- Existing compute (RTX or rented GPU) = no new hardware needed
- Active hiring in non-coding RNA = new members benefit immediately

What the agent amplifies

- Literature coverage: reads faster and more consistently than any human
- Hypothesis generation: surfaces connections across hundreds of papers
- Institutional memory: new lab members up to speed in days, not months
- Grant preparation: structures evidence and identifies gaps systematically

What the agent cannot replace

Experimental creativity and biological intuition • Wet-lab execution and troubleshooting • Peer relationships and collaboration • The judgment to know which question matters

"The era of AI-augmented biology is here. The Pasquinelli Lab has the corpus, the compute, and the questions. The only remaining step is to build the agent."